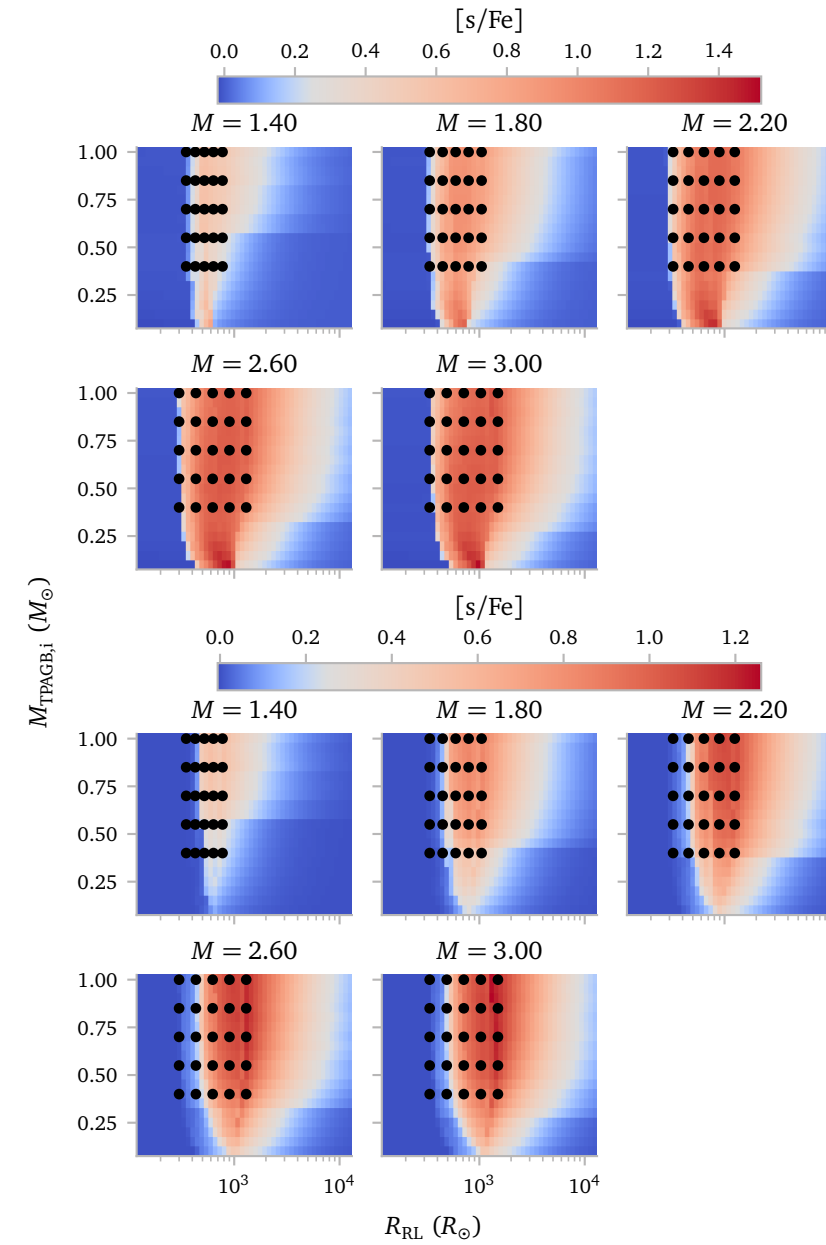


Notes Week 32

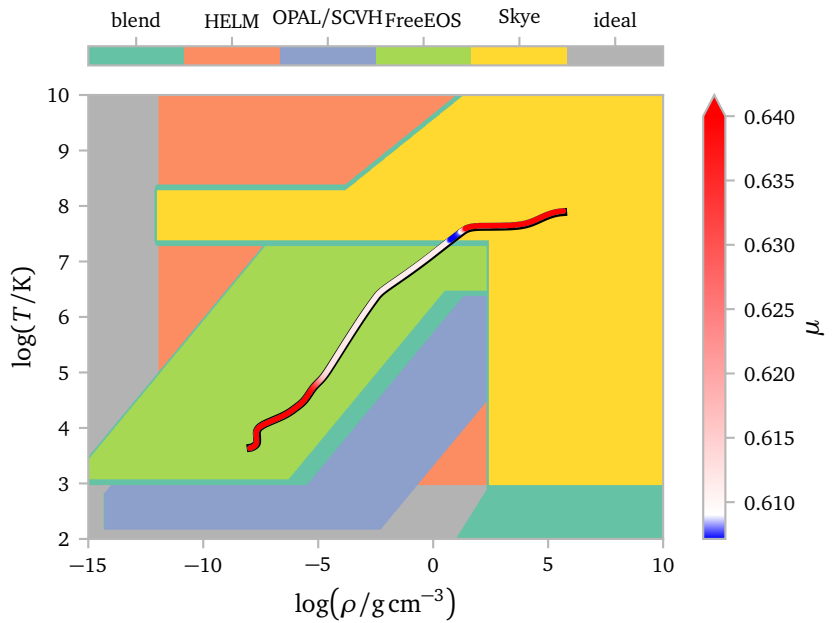
a. Grid



b. Mean Molecular Weight

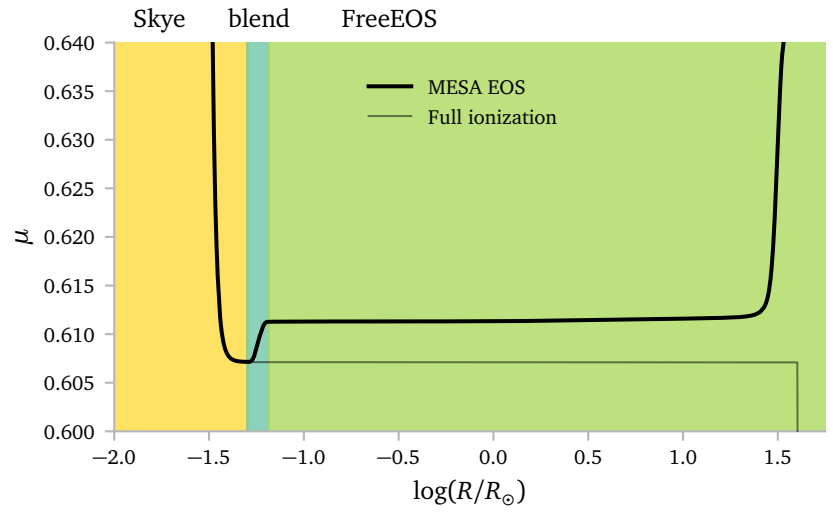
b.1 Giant Branch Dip

Using the `EOS_plotter` routine from *MESA*, it is possible to clearly plot the different EOS tables that *MESA* uses, and in which regions in the ρ - T diagram. Below, I show this, and I plot the ρ - T profile for an evolved RGB star over these tables.



The RGB profile is colored, where very red regions correspond to regions with relatively large mean molecular weight μ . Blue regions correspond to the lowest mean molecular weights, and white corresponds to an average mean molecular weight of the envelope. I think this figure quite clearly shows why we see a pit in the mean molecular weight. The core boundary happens inside the Skye EoS table (Jermyn et al., 2021). This table *assumes* complete ionization. Once the profile crosses over to the boundary region, we see that the mean molecular weight increases until it crosses the *FreeEOS* regime, which does *not assume* complete ionization.

Now that the regions of the different equations of state tables are clear, and we know that our giant branch models cross into the Skye EOS just before the core, we can again make a plot of the μ as a function of depth in the star. For the sake of clarity, I chose $\log R$ for the horizontal axis, but obviously one could chose various different quantities such as internal mass, temperature or pressure.

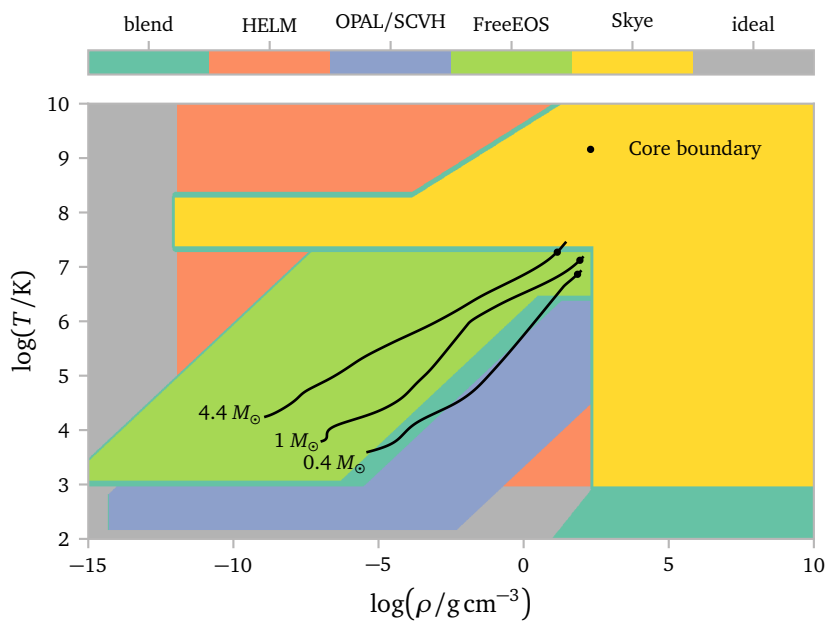


This plot *clearly* shows that the pit in the μ profile is not real. The Skye EOS (Jermyn et al., 2021) *assumes* complete ionization, which, according to the *FreeEOS*, is clearly not true up until the deepest parts of its influence. The cyan blending region just ensures the two different tables do not produce discontinuous profiles.

We also clearly see that my naive complete ionization prescription to compute the mean molecular weight perfectly matches the Skye (Jermyn et al., 2021) EOS, and starts to diverge in the blending region, just as I would expect based on the fact that Skye assumes full ionization and *FreeEOS* does not.

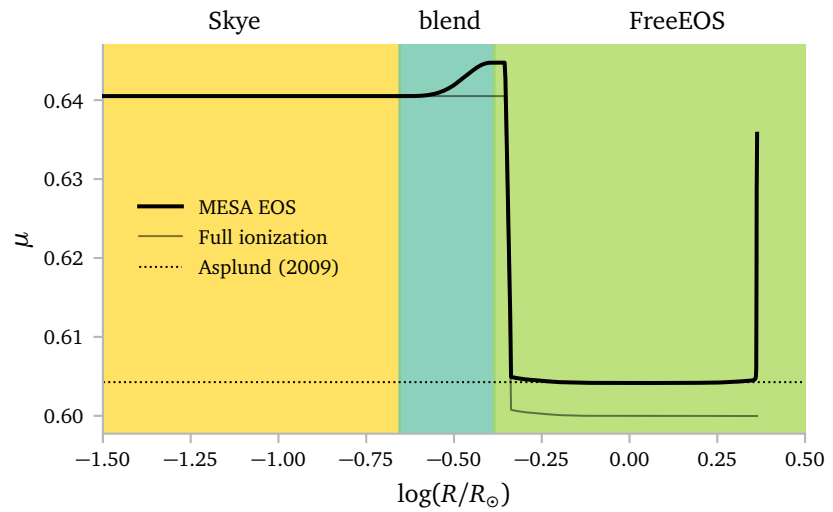
b.2 Main Sequence

Now, let's look at the main sequence. Maybe here we see a similar crossing between EOS regimes.



I once again plot the ρ - T plane and the regions of the different EOS tables. I then plot 3 different MS models with different masses and $X(H)_{\text{center}} = 0.65$. I also indicate where the hydrogen burning core boundary is with a black dot.

The plot clearly shows that the heaviest model here crosses into the Skye EOS (Jermyn et al., 2021) in its core. The entirety of the $1 M_{\odot}$ model uses the *FreeEOS* table and the $0.4 M_{\odot}$ crosses into the OPAL/SCVH region (Rogers & Nayfonov, 2002).

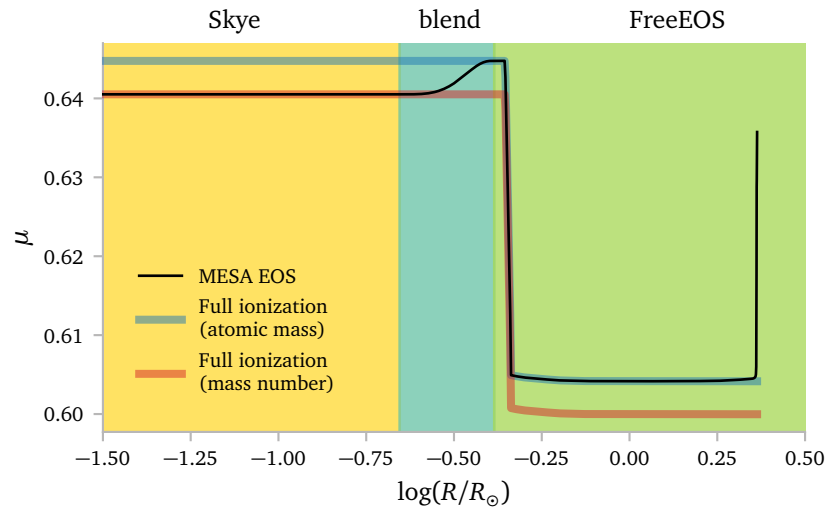


Again, we see that the mean molecular weight of the Skye EOS agrees perfectly with my fully ionized computation. As soon as we enter the blended region, the mean molecular weight starts to deviate.

The weird thing is that the Asplund et al. (2009) mean molecular weight agrees with the *FreeEOS* nearly perfectly.

It took me an embarrassingly long time, but I *think* I have finally been able to figure out what causes the differences in μ between the Skye EOS and *FreeEOS*. The differences can be explained by the masses of the elements that are used in the computation of μ . My naive approach for complete ionization for the *MESA* abundances was to simply take integer mass numbers equal to the number of protons plus neutrons. *However*, for the Asplund et al. (2009) mean molecular weight computations, I use the *Standard atomic weight*. This is a bit different.

It seems like Skye (Jermyn et al., 2021) uses the mass number, while the *FreeEOS* uses the atomic mass. In fact, when I compute the *MESA* mean molecular weight using both methods, we can clearly see that the computed mean molecular weights match depending on which mass is used.



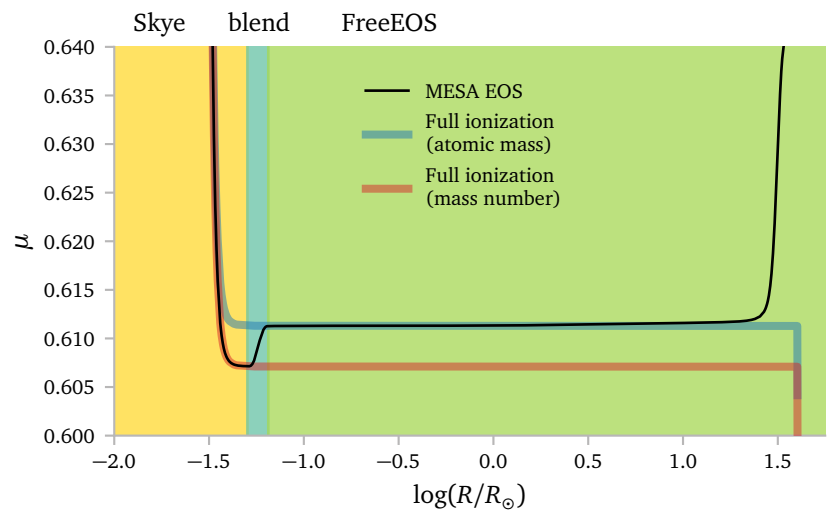
Clearly, the “mass number” mean molecular weights match the Skye (Jermyn et al., 2021) EOS mean molecular weights. The “atomic mass” mean molecular weights match the *FreeEOS* mean molecular weights.

This is not necessarily a problem. For all main sequence models, the blending region, if it even exists, is *inside* the core. This means that for the entire mixing region, the *FreeEOS* mean molecular weight is used. I use this same “atomic mass” prescription to compute μ for the accreted material, so to compute the mixing mass, we can simply use the full ionization with atomic masses. There are then no inconsistencies.

I have now implemented the fully ionized mean molecular weight computation into the accretor code. Since it agrees so well with the normal *MESA* EOS, there are no noticeable differences in the abundances or mixing masses.

b.3 Giant Branch Dip 2

I also plot the two different computation methods for the late RGB model.



Again, clearly the “mass number” mean molecular weights match the Skye (Jermyn et al., 2021) EOS mean molecular weights. The “atomic mass” mean molecular weights match the *FreeEOS* mean molecular weights.

References

Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. *ARA&A*47, 481.

Jermyn, A. S., Schwab, J., Bauer, E., Timmes, F X., & Potekhin, A. Y. (2021) *Skye: A Differentiable Equation of State*. *ApJ*913, 72.

Rogers, F. J. & Nayfonov, A. (2002) *Updated and Expanded OPAL Equation-of-State Tables: Implications for Helioseismology*. *ApJ*576, 1064.